

Manjul Tamrakar

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 Portfolio *  LinkedIn *  GitHub

Professional Summary

Full-stack mobile engineer specializing in Android media systems, SDK architecture, and production application development. Experienced in developing mobile and web products, integrating backend services through REST and GraphQL, and delivering features from prototype through production. Hands-on experience with cloud and on-device AI integration, including Gemini APIs, LiteRT-LM, and AI-assisted engineering workflows.

Experience

Veel

Mar 2026 – Present

Android Engineer

- Re-architected and rebuilt Veel's video editing SDK as a modular, maintainable Android platform using Kotlin, Jetpack Compose, Media3, Media3 Transformer, FFmpeg, JNI, and the Android NDK.
- Designed a multi-module architecture separating core models, editing operations, media processing, rendering, UI components, SDK APIs, and platform integrations.
- Developed multi-track timeline editing capabilities for video, audio, voice-over, text, filters, transformations, trimming, splitting, reordering, duplication, undo and redo.
- Implemented performant frame extraction and waveform pipelines using native FFmpeg calls through JNI, with Android media framework fallbacks for broader device compatibility.
- Built preview and export pipelines supporting multiple resolutions, frame rates, aspect ratios, audio configurations, watermarks, and gallery or URI-based output.
- Integrated GraphQL APIs, Apollo subscriptions, WebSockets, media uploads, and AI-assisted editing workflows into the Veel application.
- Improved editor performance, lifecycle handling, cancellation, memory management, media compatibility, and maintainability across complex editing workflows.

Veel

Aug 2025 – Feb 2026

Junior Android Engineer

- Developed and maintained production features for Veel's Android application using Kotlin, Jetpack Compose, Coroutines, Apollo GraphQL, Ktor, and Android platform APIs.
- Worked across product, design, backend, and mobile teams to integrate APIs, resolve lifecycle and state-management issues, and deliver reliable user-facing features.
- Diagnosed and resolved media playback, upload, networking, configuration-change, navigation, and device-compatibility issues.

COL Thinkspace

Jun 2024 – Jun 2025

Frontend Developer

- Developed mobile applications and contributed to an internal architecture for delivering full-stack web and mobile products.
- Built attendance, expense management, authentication, notification, and administrative workflows for Befit Nepal and Befit Partner.
- Led frontend development for the COL Learn learning-management platform using React, TypeScript, TanStack Query, and React Router.
- Implemented reusable components, API integration layers, application state management, and Firebase Cloud Messaging.

Mindxcape

Jul 2023 – Jun 2024

Frontend Developer

- Developed the Phase Zero Clothing e-commerce platform using Next.js, TypeScript, Zustand, NextAuth, and component-based frontend architecture.
- Integrated eSewa and Khalti payment workflows, authentication, product management, shopping cart, and checkout functionality.
- Built internal management dashboards for Boss Adventure Travels and Tours using React, Material UI, and Styled Components.

Projects

AI Translation Assistant

React Native, Expo, Gemini, LiteRT-LM

Mobile AI Application

- Built a camera-based mobile application that captures real-world text, translates it into English, and reads the translated result aloud.
- Implemented selectable cloud and on-device inference using the Gemini API and LiteRT-LM, allowing translation workflows to run through either execution path.
- Integrated Expo Camera, Expo Speech, React Native, TypeScript, and native development builds.

Befit Nepal & Befit Partner

Expo, Firebase Cloud Messaging

Mobile Application

- Developed attendance and expense management features.
- Implemented Firebase Cloud Messaging (FCM) for push notifications.

Phase Zero Clothing

Next.js, TypeScript, Zustand, NextAuth

E-commerce Platform

- Developed a production e-commerce storefront with product browsing, authentication, cart, checkout, and order-management workflows.
- Integrated Nepal-specific payment gateways including eSewa and Khalti.

Education

Tribhuvan University

2022 – 2026

B.Sc. Computer Science and Information Technology

- Patan Multiple Campus
- Maintained an average academic percentage of approximately 80

Leadership & Community

Google Developer Groups Kathmandu

Jul 2023 – Present

Lead Organizer and Technical Lead

- Organize and lead developer events including I/O Extended Kathmandu, I/O Recap Nepal, DevFest Kathmandu, workshops, technical sessions, and community programs.
- Coordinated speakers, volunteers, technical content, event operations, partnerships, and promotional campaigns.
- Contributed to the successful delivery of more than 10 technical events reaching over 800 unique participants.

Awards & Honors

SXC Sandbox Hackfest 2024

Jan 2024

Winner

- Led a team that developed an NFC-based payment platform using React Native and Firebase.

AI Crusade Hackathon 2023

Aug 2023

Winner – Environment Track

- Developed a power-grid failure prediction platform using Django and Next.js.

Technical Skills

Android	Kotlin, Jetpack Compose, Media3, ExoPlayer, FFmpeg, JNI, NDK, Coroutines, Hilt, Gradle
Mobile	Native Android, React Native, Expo, Firebase
Backend	Node.js, NestJS, Ktor, REST, GraphQL, Apollo, PostgreSQL, MongoDB
Frontend	TypeScript, React, Next.js, Astro, Tailwind CSS, Zustand, Redux
AI Tools	Gemini API, LiteRT-LM, on-device LLMs, Codex, Cursor, Claude Code